

B02 Gate A Protected Review - Logistics Shipments

Protected diagnostic review - no answer leakage to later Gates

Protected review - saved genuine Gate attempt required

Review the saved attempt, diagnose the weak control, close this PDF, complete targeted repair, then save a changed micro-proof. Gate B/C exposure still requires the existing explicit reassignment rule.

Objective fixture controls

Check	Protected control
Raw rows	120
Valid numeric rows	119
Malformed numeric	record 5038 - unit_cost=BAD
Master rows / unique keys	31 / 30
Duplicate master key	C013
Valid-row control total	86090.00
Naive dirty-master merge rows	122
Naive dirty-master merged total	87170.00
Duplicate-key inflation	1080.00

Interpretation - why a plausible pipeline can still be wrong

- Record 5038 contains a deliberately malformed numeric. If code uses errors='coerce' and silently drops the resulting null, it can manufacture a clean-looking output while hiding a source defect.
- C013 appears twice in the master. 3 valid fact rows hit that key; a naive merge inflates the control by 1080.00.
- All fact keys are present in the master, so an unmatched-key count of zero does NOT prove the merge is safe. Uniqueness/cardinality must be checked separately.
- A critical validation failure means the trusted publish path stays closed. Logs and a manifest should show the failed run and the withheld output.

Reasoning and evidence rubric

- File contract: select exactly the three intended domain files in deterministic order and record source lineage.
- Schema/numeric contract: validate required columns/types; identify the malformed value explicitly; preserve evidence rather than hiding it.
- Master contract: assert uniqueness BEFORE merge. Do not pick one duplicate master row without an approved business rule.
- Merge contract: declare many-to-one and reconcile pre/post row count plus control totals. Explain any row multiplication before aggregating.
- Validation gate: critical FAIL prevents final trusted output. A warning-only publish for a critical defect is a critical failure.
- Run evidence: log selected files, counts, checks, runtime/library versions, output status and failure/recovery path.
- Rerun evidence: a second run must not double append or silently create a second competing trusted output.
- Defense: explain input grain, output grain, why the checks are placed before publish, and what Course 3 adds later.

Critical failures

- Silent coercion/drop/dedupe that hides the injected defect.
- Merge before proving master-key uniqueness.
- Publishing a trusted output after a critical validation FAIL.
- No saved failure/recovery evidence or unsafe second-run duplication.
- Unable to explain grain, merge cardinality or validation/rerun behavior.

Repair and changed retry route

- Save the original failed/developing Gate attempt. Do not overwrite it with the repaired version.
- Open this matching protected review only after that saved attempt exists.
- Repair only the named weak competency using the targeted remediation material.
- Close this review. Produce a changed micro-proof on a different small file/key/failure condition so the repaired skill is demonstrated without copying the protected controls.
- Save the changed micro-proof plus updated validation evidence. Only then may the existing Mentor state machine allow an explicit fresh full-retest assignment when required.

Next route

PASS with all critical controls resolved: B02 ends and the course may continue to B03. FAIL/developing: targeted repair; Gate B remains sealed until the full evidence chain and explicit reassignment are satisfied.